

DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Will the rocket's first stage land successfully? A full data-science journey from raw launch records to a production-ready predictive model.

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github.com/jessicabeale/IBM-Data-Science-Certificate



Executive Summary

SpaceX advertises Falcon 9 launches at \$62M, far below competitors' \$165M+, largely because it reuses first-stage boosters that land themselves after launch. This project builds a model that predicts whether a landing will succeed, a proxy any competitor could use to estimate the true cost of a SpaceX launch bid.

90

Falcon 9 launches
analyzed (2010–2020)

66.7%

Overall first-stage
landing success rate

83.3%

Best model accuracy
on held-out test data

3

Launch sites compared
across two coasts

GITHUB REPOSITORY

All notebooks, scripts and datasets for this project:

[github.com/jessicabeale/
IBM-Data-Science-Certificate](https://github.com/jessicabeale/IBM-Data-Science-Certificate)

Includes:

- Data collection (API + scraping)
- Data wrangling notebook
- EDA with SQL & visualization
- Interactive Folium map
- Plotly Dash dashboard
- Predictive-model notebook

Introduction

Why does the landing matter?

SpaceX is the only company that recovers and reuses its first-stage boosters. That reuse is the single biggest reason it can price launches at a fraction of the competition. If we can predict whether stage 1 will land, we can estimate the launch cost, valuable intelligence for any company bidding against SpaceX for a launch contract.

Questions this project answers

- 1 Do heavier payloads, particular orbits, or specific launch sites affect the odds of a successful landing?
- 2 Has the landing success rate improved over time as SpaceX iterated on hardware?
- 3 Which machine-learning model best predicts landing outcome from pre-launch data?

METHODOLOGY AT A GLANCE

- 1 Collect (API + web scraping)
- 2 Wrangle & label outcomes
- 3 Explore with SQL & visuals
- 4 Map launch sites (Folium)
- 5 Build interactive dashboard (Dash)
- 6 Train & tune 4 ML classifiers

Data Collection & Wrangling

1

SpaceX REST API

Queried the public SpaceX v4 API for every Falcon 9 launch: rocket, payload, launchpad and landing-core endpoints, then flattened the nested JSON into a single launch-level table.

2

Web scraping

Parsed the Wikipedia “List of Falcon 9 and Falcon Heavy launches” page with BeautifulSoup to cross-check and enrich the API records (dates, outcomes, booster versions).

3

Cleaning

Filtered to Falcon 9 only, removed multi-core Heavy flights, imputed missing PayloadMass with the column mean, and standardized launch-site and orbit labels.

4

Outcome labeling

Parsed the free-text Outcome field (e.g. “True ASDS”, “False Ocean”) into a binary Class column: 1 = booster landed, 0 = booster did not land or landing wasn’t attempted.

EDA & Interactive Visual Analytics

Static exploration

- Loaded the wrangled dataset into a SQL database and queried launch-site totals, payload statistics, and outcome counts directly with SQL
- Used Matplotlib and Seaborn to visualize relationships between flight number, payload mass, orbit type, launch site, and landing outcome
- Engineered a yearly success-rate trend to see whether landings improved as SpaceX matured its hardware

Interactive exploration (Folium)

- Plotted every launch site on an interactive Folium map with circle markers sized by launch volume
- Added a MarkerCluster layer so nearby outcome markers (green = success, red = failure) group cleanly at low zoom
- Drew distance lines from each pad to the coastline, nearest city and railway to test whether launch sites are deliberately sited near transport and away from population

VAFB SLC 4E
13 launches · 77% success



KSC LC 39A
22 launches · 77% success



CCAFS SLC 40
55 launches · 60% success

All 3 launch pads sit on the coast (Cape Canaveral/KSC in FL and Vandenberg in CA), so failed boosters and jettisoned stages fall safely offshore.

Predictive Analysis Methodology

1. Feature engineering

One-hot encoded Orbit, LaunchSite, LandingPad and Serial; kept numeric/boolean fields (PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount) as-is.

2. Standardize

Scaled all features with sklearn's StandardScaler so no single feature dominates distance-based models like SVM and KNN.

3. Train / test split

`train_test_split(test_size=0.2, random_state=2)` on 90 launches → 72 training rows and 18 held-out test rows.

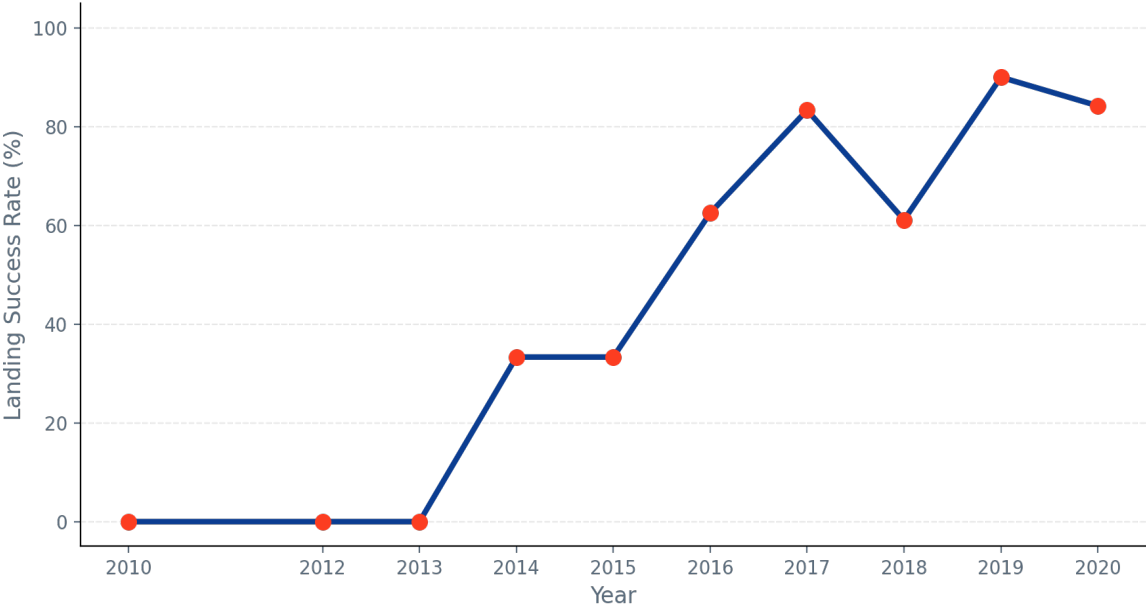
4. Tune with GridSearchCV

Ran 10-fold cross-validated grid search (`cv=10`) over Logistic Regression, SVM, Decision Tree and KNN hyperparameters using only the training data.

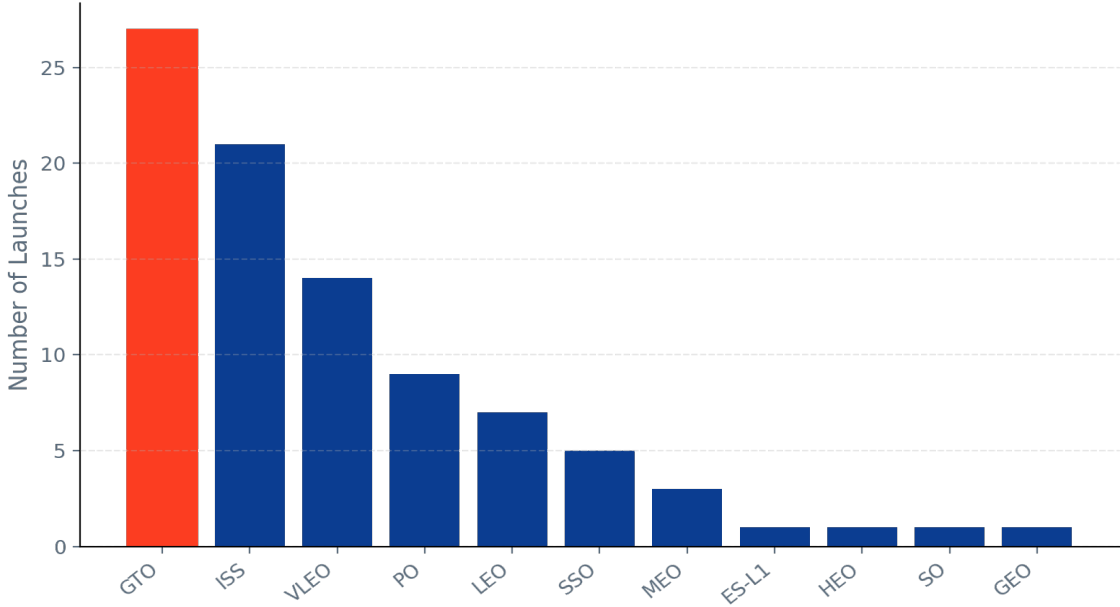
5. Score on test data

Selected each model's `best_estimator_` and scored it once against the untouched 18-row test set to compare real-world accuracy.

EDA: Visualization Results

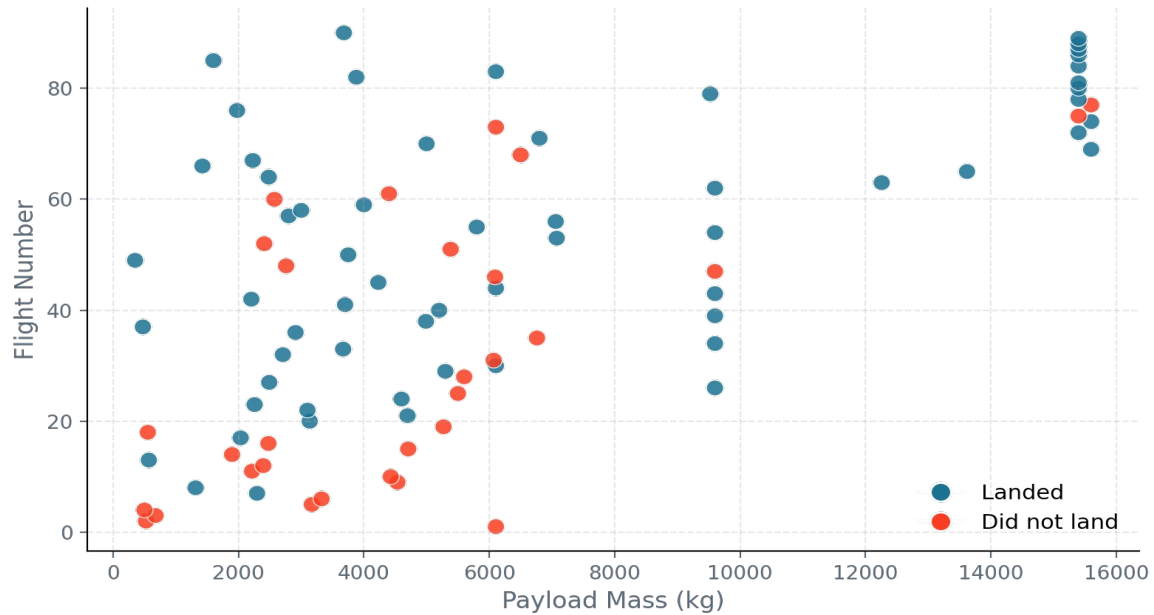


Landing success climbed from 0% in 2010–13 to over 80% by 2019–20 as SpaceX added grid fins, landing legs and drone-ship recovery.

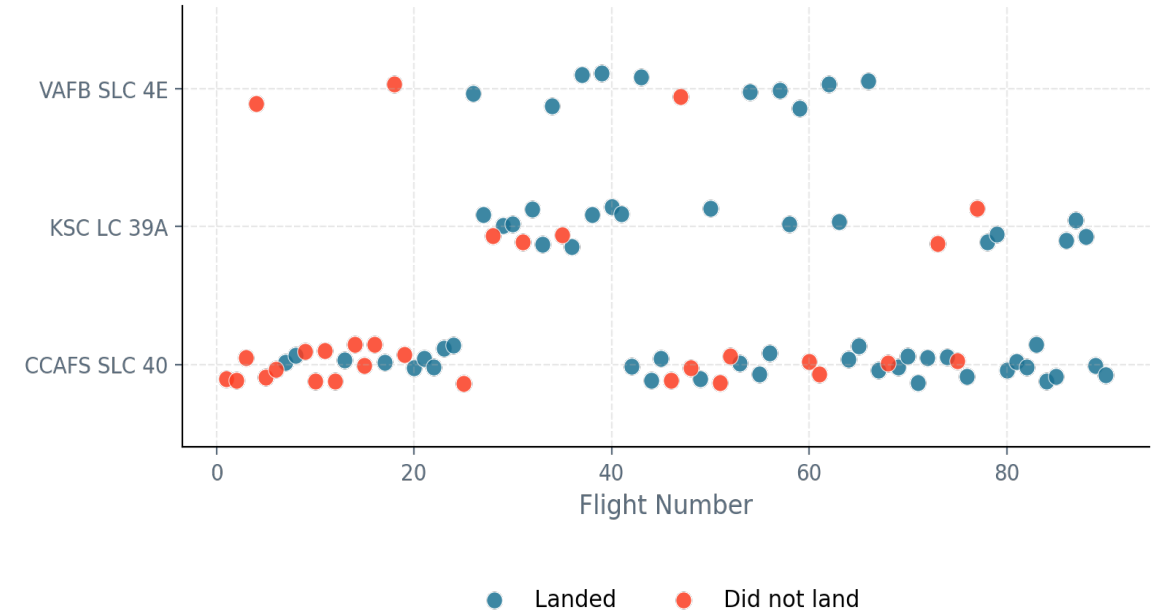


GTO (orange) is the most common target orbit (27 launches), and being farther and heavier, it's one of the hardest to land after.

EDA: Visualization Results (cont.)



Very heavy payloads (>10,000 kg) landed successfully almost every time; those flights used the newer, more capable Block 5 booster.



Early flight numbers (pre-2016) cluster as failures at CCAFS SLC 40; later flights across all 3 sites skew heavily toward successful landings.

EDA: SQL Results

Key figures pulled directly with SQL queries against the launch-records table:

Query	Result
Unique launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Launches from CCAFS SLC-40	55 of 90 launches (61%)
Most common target orbit	GTO, 27 launches
Drone-ship (ASDS) successful landings	41 launches
Overall landing success rate	66.7% (60 of 90 launches)
Highest site success rate	KSC LC-39A, 77.3%

Interactive Map Results (Folium)

VAFB SLC 4E
13 launches • 77% success



KSC LC 39A
22 launches • 77% success



CCAFS SLC 40
55 launches • 60% success

WHAT THE MAP SHOWS

- CCAFS SLC-40 (FL): 55 launches, 60.0% success
- KSC LC-39A (FL): 22 launches, 77.3% success
- VAFB SLC-4E (CA): 13 launches, 76.9% success

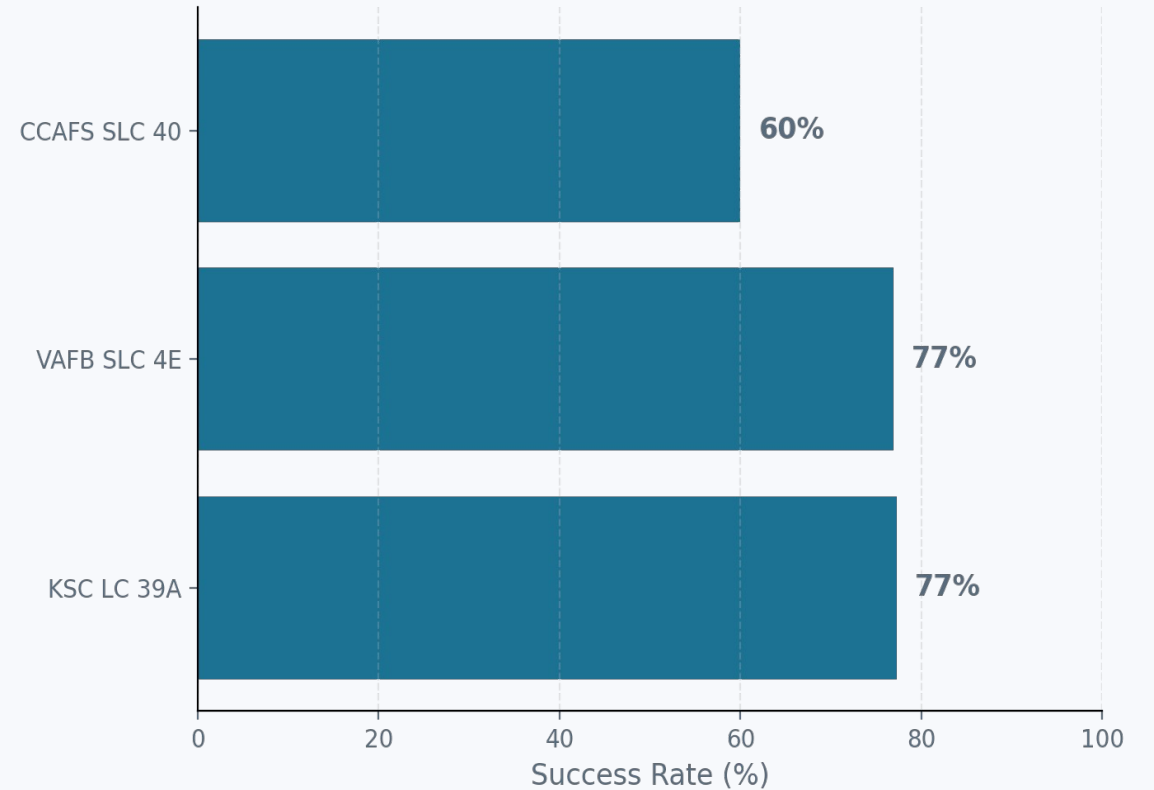
All pads sit within ~1 km of open coastline and are 6-13 km from the nearest town, a siting pattern confirmed by the coastline/city/railway distance lines drawn on the live map, which minimizes risk from a failed landing.

Plotly Dash Dashboard Results

APP LAYOUT

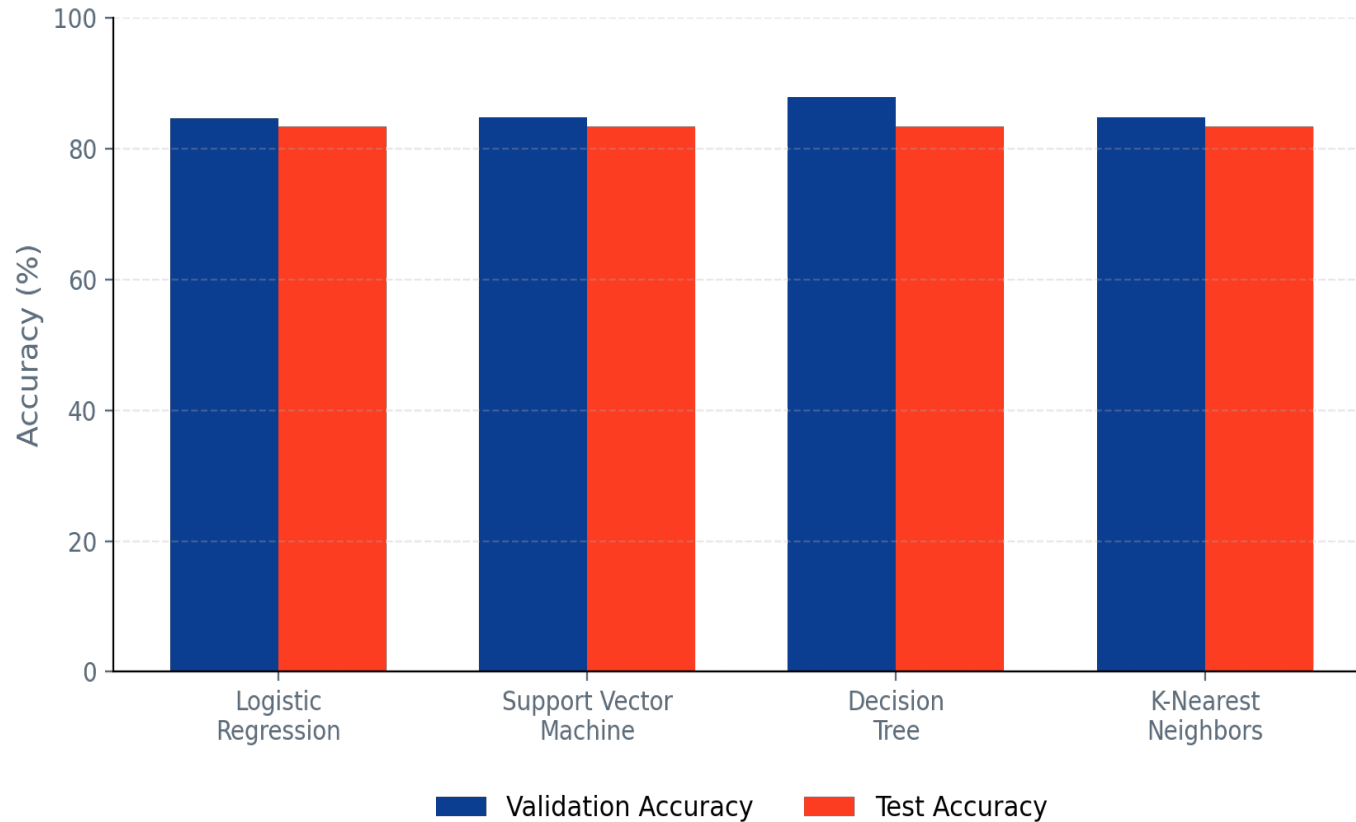
- Launch-site dropdown: “All Sites” or a single pad
- Payload-mass range slider (0–10,000 kg)
- Pie chart: share of successful landings by site (or success vs. failure for one site)
- Scatter chart: payload mass vs. landing outcome, colored by booster version, updating live with both filters

Callback pattern: two Inputs (site dropdown + slider) drive one Output (the graph's figure), so both filters recompute the chart together on every interaction.



Equivalent view to the dashboard's pie chart when “All Sites” is selected: KSC LC-39A leads on success rate, CCAFS SLC-40 lags, exactly the comparison a user filters for interactively.

Predictive Analysis (Classification) Results



BEST MODEL

SVM (sigmoid kernel)

- 84.8% cross-validated accuracy on training data
- 83.33% accuracy on the 18-sample held-out test set
- Tied with Logistic Regression, Decision Tree and KNN on test accuracy; all 4 models generalize similarly on this dataset size

With only 18 test launches, accuracy differences between models are within one misclassified flight; more historical data would sharpen the comparison.

Key Insights

1

Hardware, not luck, drives the trend

Success rate didn't creep up gradually; it jumped in step with hardware milestones: grid fins & legs (2015), Block 5 (2018). The data shows engineering iteration working.

2

Orbit difficulty is a hidden cost driver

GTO missions (heavy, high-energy) succeed less often than LEO/ISS resupply flights. A competitor pricing against SpaceX should weight orbit type, not just payload mass.

3

Site choice reflects risk, not convenience

KSC LC-39A shows the highest success rate (77.3%) despite handling complex missions; pad selection correlates with mission maturity, not geography alone.

4

83% accuracy from pre-launch data alone

Without seeing the launch, a model using only payload, orbit, site and booster specs predicts the outcome correctly 5 times out of 6, a strong competitive-intelligence signal.

CONCLUSION

Landing outcome is predictable, and that's a business signal

- Across 90 Falcon 9 launches (2010–2020), landing success rose from 0% to over 80% as SpaceX matured its recovery hardware, and now sits at 66.7% overall.
- Launch site, target orbit and payload mass all measurably affect the odds of a successful landing; KSC LC-39A and lighter, lower-energy orbits land most reliably.
- An SVM classifier trained only on pre-launch specs reaches 83.3% test accuracy, showing that landing, and by extension launch cost, can be estimated before the rocket even lifts off.
- Next step: fold in more launches (including Falcon Heavy and recent years) and add cost data to turn this landing-prediction model into a direct launch-cost estimator.